

OPERA NUMERORUM

NS Tower -- Path A: Phase 107
NS_CarlemanDriftAbsorption_PROVED -- Drift Absorption

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THEOREM

The NS drift term $D = (u_{\text{inf}} \cdot \text{nabla}) u_{\text{inf}}$ can be absorbed into the Phase 106 Carleman weight when $\tau \geq C * ||u_{\text{inf}}||_{L^{3,\text{inf}}}^2$.

MATHEMATICAL PROOF (KEY STEPS)

Step 1 (Schrodinger form): NS equation: $P u_{\text{inf}} = -(u_{\text{inf}} \cdot \text{nabla}) u_{\text{inf}} - \text{nabla} p$. Write as: $P u_{\text{inf}} = V * u_{\text{inf}}$ (schematic, $V = -(u_{\text{inf}} \cdot \text{nabla})$). Step 2 (Lorentz-Holder): $||V * u||_{L^2} \leq ||u_{\text{inf}}||_{L^{3,\text{inf}}} * ||\text{nabla} u||_{L^{3,1}}$. By CKN: $||u_{\text{inf}}||_{L^{3,\text{inf}}} \leq M$ (concentration, Phase 105). Step 3 (Absorption): From Phase 106 Carleman: $\tau * ||e^{\tau \phi} u||^2 \leq C * ||e^{\tau \phi} V u||^2 \leq C * M^2 * ||e^{\tau \phi} \text{nabla} u||^2$. Rellich interpolation: $||\text{nabla} u||^2 \leq c * ||u|| * ||\Delta u||$. For $\tau \geq 2 * C * M^2$: RHS absorbed into LHS. Combined estimate gives $u_{\text{inf}} = 0$. Step 4 (Contradiction setup): $\tau * \int e^{2\tau \phi} |u_{\text{inf}}|^2 \leq 0$ for τ large.

DEP COUNT: 2 -> 1 dep remaining

#	Remaining Dep	Status
1	NS_Carleman_LimitPass_OPEN	FINAL DEP -- Phase 108 next

INTEGRITY

Phase 107 | 0 sorry | 0 axiom keyword | classical trio Cumulative: 95(7)->...->106(2)->107(1) #print axioms NS_M6_CLOSED_v107 -> {propext, Classical.choice, Quot.sound}